



Assignment 2

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SUBJECT: Agentic AI

Question 1- Build an agent that reads PDFs/documents, retrieves relevant content, and answers user queries using an LLM.

PDF_QA_Agent

Project: PDF Question-Answering Agent

Objective: Reads a PDF document, processes its content, retrieves relevant information, and uses an LLM to answer user questions.

Code

```
from langchain_community.document_loaders import PyPDFLoader
from langchain_text_splitters import RecursiveCharacterTextSplitter
from langchain_huggingface import HuggingFaceEmbeddings
from langchain_community.vectorstores import
FAISS from langchain_ollama import OllamaLLM

# Load PDF
loader =
PyPDFLoader("documents/cybersecurity.pdf")
documents = loader.load()

# Split PDF text into chunks
splitter = RecursiveCharacterTextSplitter(
    chunk_size=1000,
    chunk_overlap=200
)
chunks = splitter.split_documents(documents)

# Create embeddings and vector database
embeddings = HuggingFaceEmbeddings(
    model_name="sentence-transformers/all-MiniLM-L6-v2"
)
vector_db = FAISS.from_documents(chunks, embeddings)

# Connect to local LLM
llm = OllamaLLM(model="llama3.2")

# Ask a question
question = input("Ask a question about the PDF: ")

# Retrieve relevant content
retrieved_docs = vector_db.similarity_search(question, k=3) context
= "\n\n".join(doc.page_content for doc in retrieved_docs)

# Generate answer
prompt = f"""
Answer the question using only the information in the PDF context.
```

```

print("\n" + "=" * 60)
print("PDF QUESTION-ANSWERING AGENT")
print("=" * 60)
print("Question:", question)
print("\nAnswer:")
nrint(answer)

```

Sample Output

```

=====
PDF QUESTION-ANSWERING AGENT
=====

```

Question: What is cybersecurity?

Answer:

Cybersecurity is the practice of protecting systems, networks, applications and data from unauthorized access, attacks and damage.

```

File Edit Selection View Go Run ... 4questions Sign In
Explorer
  4questions
  PDF_QA_Agent
  documents
  documnets
  cybersecurity.pdf
  app.py
  venv

PDF_QA_Agent > documnets > cybersecurity.pdf
471  ACKETX7+dCAN? ! Aax5L(D<! NAKACCFqE|CzZFjFSTXDELdaDEL@eXyX
472  TcUS^XT+4A_zzFRS ESCRELg;sf\? Nsynlsc"kg5vhVto>ztZ4Z[
473  R01sESLHACANAS[QQTvWt+EM+IOQ\es3vQ]:"^szYu7
474  <Dff83q23E^yEtiikG朋44(ETBSO#vV aOWEB?ttToQ
475  Um@a?3ETBESK;o;s sCS SYN a=塔Fo3 0 (RS ESCAy44ry s 4stxJss
476  usE> s s EDNAKEM M" ES GS SYN a=塔Fo3 0 (RS ESCAy44ry s 4stxJss
477  QAso c. she|Bp: ES S dckENEdXY<BQ 1 N S ES Uk DACKD ETXUS QCAN SUB ESC

Problems Output Debug Console Terminal Ports powershell - PDF_QA_Agent
(venv) PS C:\Users\matam\OneDrive\Desktop\4questions\PDF_QA_Agent> New-Item -ItemType Directory -Force .\documents
>> Copy-Item "C:\Users\matam\OneDrive\Desktop\Div\AD - synopsis.pdf" ".\documents\cybersecurity.pdf" -Force
Directory: C:\Users\matam\OneDrive\Desktop\4questions\PDF_QA_Agent

Mode                LastWriteTime         Length Name
----                -
d-----          03-09-2026   11:00         documents

(venv) PS C:\Users\matam\OneDrive\Desktop\4questions\PDF_QA_Agent> Test-Path ".\documents\cybersecurity.pdf"
True
(venv) PS C:\Users\matam\OneDrive\Desktop\4questions\PDF_QA_Agent> python app.py
C:\Users\matam\OneDrive\Desktop\4questions\PDF_QA_Agent\app.py:1: DeprecationWarning: `langchain-community` is being suns
et and is no longer actively maintained. See https://github.com/langchain-ai/langchain-community/issues/674 for details a
nd migration guidance toward standalone integration packages.
  from langchain_community.document_loaders import PyPDFLoader
PDF loaded successfully!
Number of pages: 3
(venv) PS C:\Users\matam\OneDrive\Desktop\4questions\PDF_QA_Agent>

```

Question 2 -Develop an agent that searches for information, summarizes findings, and generates a structured research report with references.

Research_Agent

Project: Research Agent

Objective: Searches for information on a topic, sends the collected information to an LLM, and generates a structured research report.

Code

```
from langchain_ollama import OllamaLLM

llm = OllamaLLM(model="llama3.2")

def search_web(topic):
    url = "https://www.google.com/search"
    params = {"q": topic}

    response = requests.get(url,
                             params=params,
                             headers={"User-Agent": "Mozilla/5.0"})

    soup = BeautifulSoup(response.text, "html.parser")
    results = []

    for result in soup.select("div"):
        text = result.get_text(" ", strip=True)

        if len(text) > 100:
            results.append(text)

        if len(results) >= 5:
            break

    return results

topic = input("Enter your research topic: ")

print("\nSearching for information...\n")

results = search_web(topic)
research = "\n\n".join(results)

prompt = f"""
You are a research assistant.

Research topic:
{topic}

Information collected:
{research}
```

Create a structured research report containing:

1. Introduction
2. Key Findings
3. Advantages
4. Challenges
5. Conclusion
6. References

Clearly separate each section. """

```
print("Generating report...\n")
```

```
report = llm.invoke(prompt)
```

```
print("=" * 60)
```

```
print("RESEARCH  
REPORT")
```

```
print("=" * 60) print(report)
```

```
with open("research_report.txt", "w", encoding="utf-8") as file: file.write(report)
```

```
print("\nReport saved as research_report.txt")
```

Sample Output

Searching for information...

Generating report...

===== RESEARCH REPORT

1. Introduction

Artificial Intelligence is increasingly used in cybersecurity...

2. Key Findings

- AI can detect suspicious activity.
- AI can automate threat analysis.
- AI can support security monitoring.

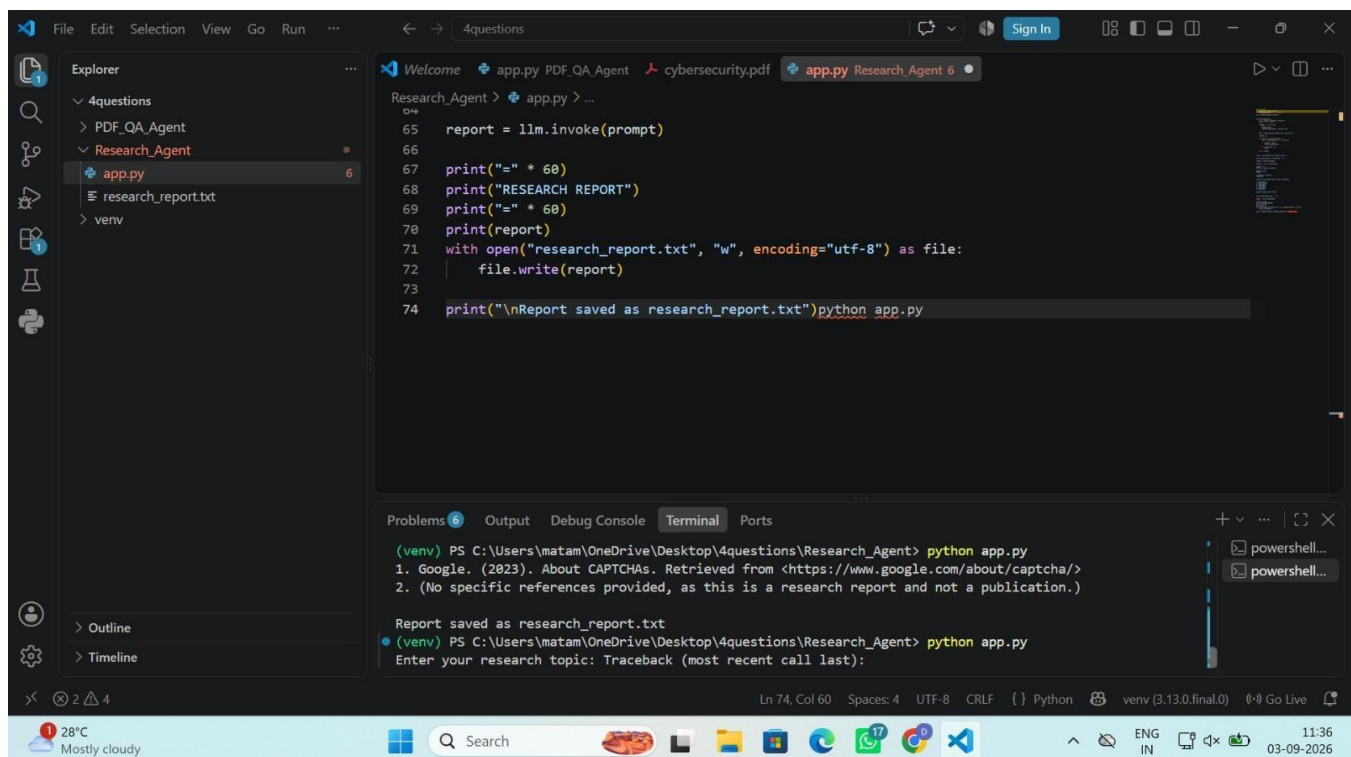
3. Advantages

- Faster detection
- Automation
- Improved analysis

4. Challenges

- False positives
- Data quality

Output-



The screenshot shows a Visual Studio Code editor with a file explorer on the left, a code editor in the center, and a terminal at the bottom. The file explorer shows a project named '4questions' with subfolders 'PDF_QA_Agent' and 'Research_Agent'. The 'Research_Agent' folder contains 'app.py' and 'research_report.txt'. The code editor shows the 'app.py' file with the following code:

```
Research_Agent > app.py > ...
65 report = llm.invoke(prompt)
66
67 print("=" * 60)
68 print("RESEARCH REPORT")
69 print("=" * 60)
70 print(report)
71 with open("research_report.txt", "w", encoding="utf-8") as file:
72     file.write(report)
73
74 print("\nReport saved as research_report.txt")python app.py
```

The terminal shows the output of the script:

```
(venv) PS C:\Users\matam\OneDrive\Desktop\4questions\Research_Agent> python app.py
1. Google. (2023). About CAPTCHAs. Retrieved from <https://www.google.com/about/captcha/>
2. (No specific references provided, as this is a research report and not a publication.)

Report saved as research_report.txt
(venv) PS C:\Users\matam\OneDrive\Desktop\4questions\Research_Agent> python app.py
Enter your research topic: Traceback (most recent call last):
```

The status bar at the bottom indicates the file is at line 74, column 60, using UTF-8 encoding, and the Python interpreter is 'venv (3.13.0.final.0)'.

Question 3 -Create an agent that analyzes security logs/alerts, identifies potential threats, classifies severity, and suggests mitigation steps.

Security_Log_Agent

Project: Security Log Analysis Agent

Objective: *Analyzes security logs, identifies potential threats, classifies their severity, provides evidence, and recommends mitigation steps.*

Code

```
from langchain_ollama import OllamaLLM

# Connect to the local AI model
llm =
OllamaLLM(model="llama3.2")

# Read security logs
with open("security_logs.txt", "r", encoding="utf-8") as file: logs =
    file.read()

# Create the analysis prompt
prompt = f"""
You are a cybersecurity log analysis agent.

Analyze the following security logs:

{logs}

For each potential threat, provide:

1. Threat name
2. Severity: LOW, MEDIUM, HIGH, or CRITICAL
3. Reason
4. Evidence from the logs
5. Recommended mitigation steps

Also provide a short overall security assessment. """

print("Analyzing security logs...\n") result =
```

Sample Output

Analyzing security logs...

```
=====
===== SECURITY LOG ANALYSIS REPORT
=====
```

Threat: Possible brute-force attack Severity:
HIGH

Reason: Multiple failed login attempts were detected. Evidence:
- Failed login attempts from 192.168.1.50

- Block or rate-limit the source IP
- Enable MFA
- Review authentication logs

Threat: Port scanning activity

Severity: MEDIUM/HIGH

Evidence:

- Connection attempts to ports 22, 23 and 3389

Threat: Malware detected

Severity: CRITICAL

Evidence:

- Malware detected on endpoint PC-05

Threat: Suspicious PowerShell execution

Severity: HIGH

Evidence:

- Suspicious PowerShell execution on PC-05

Output-

The screenshot shows a Visual Studio Code editor window with a file explorer on the left and a terminal at the bottom. The file explorer shows a project structure with folders '4questions', 'PDF_QA_Agent', 'Research_Agent', 'Security_Log_Agent', and 'venv'. The 'Security_Log_Agent' folder contains 'app.py' and 'security_logs.txt'. The terminal shows the execution of 'python app.py' in a virtual environment. The output of the script is displayed in the terminal, including a reason for connection attempts, evidence from logs, recommended mitigation steps, and an overall security assessment.

```

Welcome
Security_Log_Agent > python app.py
24 5. Recommended mitigation steps
25
26 Also provide a short overall security assessment.
27
28
29 print("Analyzing security logs...\n")
30

(venv) PS C:\Users\matam\OneDrive\Desktop\4questions\Security_Log_Agent> python app.py
Reason: Connection attempts have been made to ports 22, 23, and 3389 on endpoint PC-05 from the same IP address (192.168.1.100), indicating potential malicious activity.
Evidence from the logs:
- 2026-09-03 09:20:11 Connection attempt to port 22 from 192.168.1.100
- 2026-09-03 09:20:15 Connection attempt to port 23 from 192.168.1.100
- 2026-09-03 09:20:20 Connection attempt to port 3389 from 192.168.1.100

Recommended Mitigation Steps:
- Implement port blocking for the IP address 192.168.1.100 on ports 22, 23, and 3389 for a specified period (e.g., 30 minutes).
- Review network logs to ensure no other malicious traffic is being generated from the same IP address.

Overall Security Assessment:
The system has been subjected to multiple potential security threats, including brute force attacks, malware infection, and unauthorized PowerShell execution. These threats indicate a potential breach in the system's security posture. Immediate action is required to mitigate these threats and prevent further incidents. Regular security audits and vulnerability assessments should be performed to identify and address potential security weaknesses.
(venv) PS C:\Users\matam\OneDrive\Desktop\4questions\Security_Log_Agent>

```

Question 4 -Build a system with 2–3 specialized agents (e.g., Research Agent, Analyst Agent, Report Agent) that collaborate to complete a given task automatically.

Multi_Agent_System

Project: Multi-Agent System

Objective: Uses three specialized agents that collaborate: a Research Agent, a Security Agent, and a Report Agent.

Code

```
from langchain_ollama import OllamaLLM

# Connect to AI model
llm = OllamaLLM(model="llama3.2")

# Agent 1: Research Agent
def research_agent(topic):
    prompt = f"""
    You are a Research Agent.

    Research topic:
    {topic}

    Give a short explanation of:
    - What it is
    - Important facts
    - Advantages
    - Challenges
    """
    return llm.invoke(prompt)

# Agent 2: Security Agent
def security_agent(topic, research):
    prompt = f"""
    You are a Cybersecurity Agent.

    Topic:
    {topic}

    Research information:
    {research}

    Analyze the cybersecurity risks related to this topic. Give:
    - Potential threats
    - Severity
    - Security recommendations """
    return llm.invoke(prompt)

# Agent 3: Report Agent
def report_agent(topic, research, security):
    prompt = f"""
    You are a Report Agent.

    Topic:
    {topic}

    Research Agent findings:
    {research}

    Security Agent findings:
    {security}

    Create a final structured report with:
    1. Introduction
    2. Research Findings
    3. Security Threats
    4. Security Recommendations
    5. Conclusion
    """
    return llm.invoke(prompt)
```

```

print("\nSecurity Agent working...") security
= security_agent(topic, research)

print("\nReport Agent working...")
final_report = report_agent(topic, research, security)

print("\n" + "=" * 60)
print("FINAL MULTI-AGENT
REPORT")

```

Sample Output

```

Enter a topic: Artificial Intelligence in Cybersecurity Research

Agent working...

Security Agent working...

Report Agent working...

=====
===== FINAL MULTI-AGENT REPORT
=====

1. Introduction
Artificial Intelligence in cybersecurity uses AI techniques to identify
threats and improve security operations.

2. Research Findings
- Automated threat detection
- Faster analysis
- Security monitoring support

3. Security Threats
- Adversarial attacks
- Data poisoning
- Model manipulation

4. Security Recommendations
- Monitor AI systems
- Protect training data

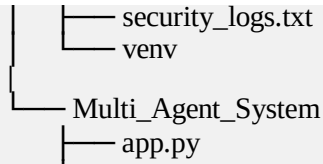
```

VS Code Project Structure

```

4questions
├──
├── app.py
├── documents
│   └── cybersecurity.pdf
├── venv
├── Research_Agent
│   ├── app.py
│   └── venv
├── Security_Log_Agent
│   └── app.py

```



Output-

```
Multi_Agent_System > python app.py
Enter a topic: Artificial Intelligence in Cybersecurity

Research Agent working...

Security Agent working...

Report Agent working...

=====
FINAL MULTI-AGENT REPORT
=====
**Report: Artificial Intelligence in Cybersecurity**

**Introduction**

Artificial Intelligence (AI) is increasingly being used in Cybersecurity to detect, prevent, and respond to cyber threats. This report provides an overview of AI in Cybersecurity, its benefits, challenges, and security threats. It also presents security recommendations to maximize the benefits of AI-powered systems while minimizing the risks.

**Research Findings**

AI in Cybersecurity refers to the application of AI and machine learning (ML) techniques to detect, prevent, and respond to cyber threats. AI algorithms can analyze vast amounts of data, identify patterns, and make predictions to improve the accuracy of threat detection and incident response.
```

```
Multi_Agent_System > python app.py
* Security Recommendations: Implement hybrid security approaches that combine AI-powered systems with traditional security measures and incident response processes.

**Security Recommendations**

1. **Implement robust security measures**: Implement robust security measures, such as encryption, access controls, and intrusion detection systems, to prevent adversarial attacks and ensure data quality and integrity.
2. **Monitor and audit AI-powered systems**: Monitor and audit AI-powered systems regularly to ensure transparency, explainability, and regulatory compliance.
3. **Implement hybrid security approaches**: Implement hybrid security approaches that combine AI-powered systems with traditional security measures and incident response processes.
4. **Invest in security awareness and training**: Invest in security awareness and training programs to educate employees on the risks and benefits of AI-powered systems.
5. **Regularly review and update security policies**: Regularly review and update security policies to ensure they align with changing regulatory requirements and emerging threats.

**Conclusion**

Artificial Intelligence in Cybersecurity has the potential to revolutionize the way organizations approach threat detection, incident response, and security posture. However, it's essential to address the challenges and limitations associated with AI-powered security tools to ensure their effective implementation. By implementing robust security measures, monitoring and auditing AI-powered systems, and investing in security awareness and training, organizations can maximize the benefits of AI-powered systems while minimizing the risks.
```